

Learning objectives

In this lesson you will learn to:

- Give reasons why computers are connected on a network
- Differentiate between a LAN and a WAN
- Categorise tasks according to the type of network required to carry them out
- Explain the benefits to organisations of a WAN
- Explain why protocols are needed on a network
- Describe the purpose of an IP address.

For more information and additional learner activities see Chapter 4.1 of the student book.

How useful is a stand-alone computer?

Imagine you have a computer that is not connected to any others.

It has no network connection to a local network, the internet or other devices.

What could you use this device for?

Would it be useful in today's world?

This kind of computer is called a **stand-alone computer** because it 'stands alone', without connections to other computers or networks.

Think about computer networks

- How are devices connected to each other?
- What benefits are there to using a connected computer?
- What tools/applications do you have access to through a computer network?
- What are some of the things you could not do if your computer was not connected to another device?

Categorising a network

Networks can be classified based on their geographical coverage.

Networks that are contained within a single building or site are known as **Local Area Networks (LANs)**.

Networks that are connected over a larger geographical area are known as **Wide Area Networks (WANs)**.

What type of network am I using?

Think of some examples of tasks you complete using a computer.

Most of these will involve a network of some kind.

For example: Printing a document requires the use of a LAN. The computer will be connected to a printer either directly or through the school's network.

Why organisations have WANs

Big organisations may have their own WAN.

The organisation owns all of the WAN's hardware and has full control of the network. Others cannot use its infrastructure.

This means the network can be very secure.

This is a very expensive option because the infrastructure has to be installed and maintained.

Benefits of networks

Users can:

- ✓ share data and information with each other
- ✓ communicate with each other using email, messaging, video conferencing, etc.
- ✓ work collaboratively using web-based software, virtual meeting tools, etc.
- ✓ share hardware devices such as printers
- ✓ make use of centralised support and maintenance
- ✓ socialise with one another

The internet

The internet is the largest WAN in the world.
It is a global network of networks.

There are potentially billions of devices connected to the internet at any one time.

Can you foresee some potential problems with this?

Connecting globally

In order for multiple different devices to speak to each other, they need to communicate in the same way.

To do this, devices use **protocols**.

Protocols are sets of rules (agreed standards) that govern the communication between devices.

For two devices to communicate, they must use the same protocol.

There are number of protocols you need to know about and we will look at them in more detail in later lessons.

Finding the device you need

Another key element needed for communication is an addressing system.

Each device in a network needs a unique address so that connections can be made to it.

There are two types of address that you will learn about.

- IP addresses are used on the internet.
- MAC addresses are used on LANs.

IP Addresses

In a WAN, an IP (Internet Protocol) address is used to identify an individual device or network access point.

There are two types of IP address. These are known as version 4 and version 6 (IPv4, IPv6) addresses.

IPv6 addresses are longer. IPv6 can therefore address a greater number of unique devices. They were created to help manage the huge number of devices on the internet.

Wrap up: you have learned how to...

- ✓ Give reasons why computers are connected on a network.
 - Sharing resources and devices.
- ✓ Differentiate between a LAN and a WAN.
 - Local Area Networks are contained within a single site.
 - Wide Area Networks cover a larger geographical area.
- ✓ Explain the benefits an organisation gets from using a WAN.
 - Able to share data and resources between different sites.
 - Owning the network hardware gives more control. A WAN can be more secure than the VPN alternative.
- ✓ Explain why protocols are needed on a network.
 - Protocols mean that devices are able to 'speak the same language'.
- ✓ Describe the purpose of an IP address.
 - Used to identify a device or network access point on the internet.